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Problem Set 3, COSC330/PHYS306
Chapters 3 and 4

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1 Chapter 3

1. **[Section 3-9, Exercise 39]** Examine the conditions indicated in Figure 3–92, and identify the faulty gates.

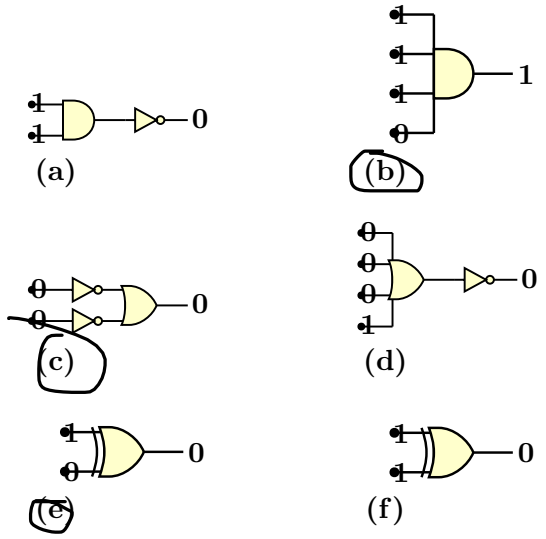


FIGURE 3–92

2. **[Section 3-9, Exercise 40]** Determine the faulty gates in Figure 3–93 by analyzing the timing diagrams.

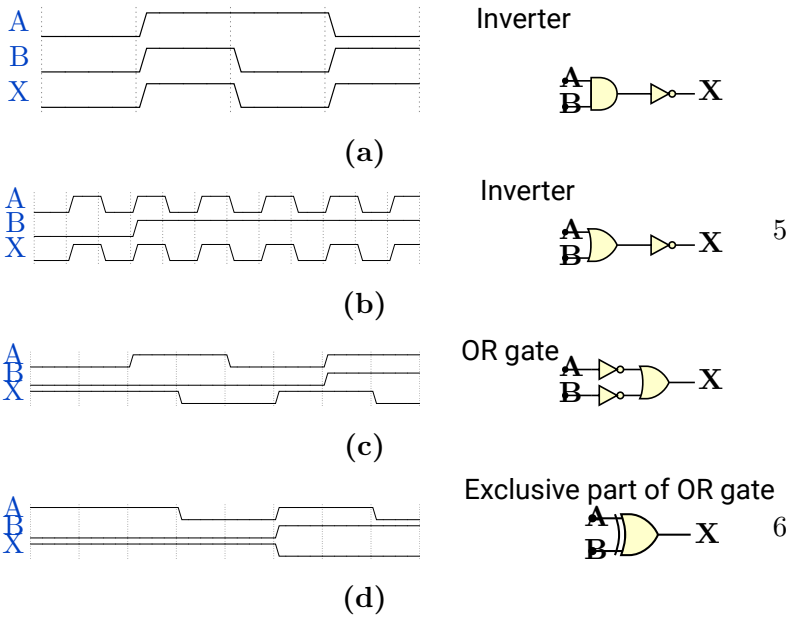


FIGURE 3–93

2 Chapter 4

3. **[Section 4-3, Exercise 9]** Apply DeMorgan's theorems to each expression:

(a) $\overline{A+B}$ = $\overline{A} * \overline{B}$

(b) $\overline{\overline{A}B}$ = $A + \overline{B}$

(c) $\overline{A+B+C}$ = $\overline{A} * \overline{B} * \overline{C}$

(d) $\overline{ABC}$ = $\overline{A} + \overline{B} + \overline{C}$

(e) $\overline{A(B+C)}$ = $\overline{A} + \overline{B}*\overline{C}$

(f) $\overline{AB+CD}$ = $\overline{A} + \overline{B} * \overline{C} + \overline{D}$

(g) $\overline{\overline{A}B+CD}$ = $A+\overline{B} * \overline{C}+\overline{D}$

(h) $\overline{(A+\overline{B})(\overline{C}+D)}$ = $(\overline{A}*B) + (C*\overline{D})$

4. **[Section 4-4, Exercise 13]** Write the Boolean expression for each of the logic circuits in Figure 4–57.

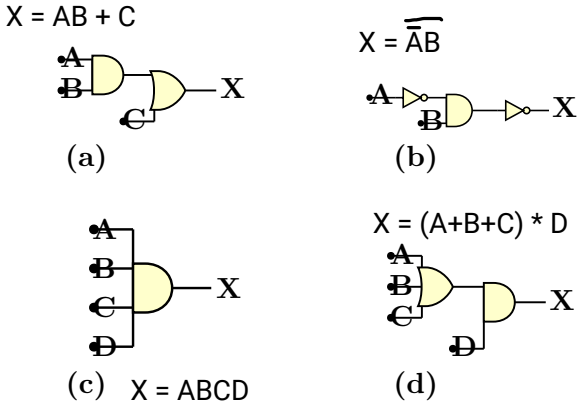


FIGURE 4–57

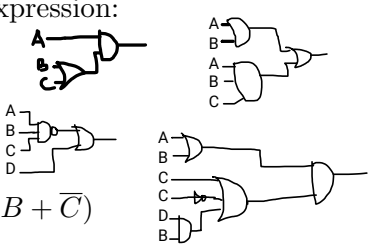
5. **[Section 4-4, Exercise 15]** Draw the logic circuit represented by each expression:

(a) $A(B+C)$

(b) $AB+ABC$

(c) $\overline{ABC}+D$

(d) $A+B(C+DB+\overline{C})$



6. **[Section 4-4, Exercise 18]** Construct a truth table for each of the following Boolean expressions:

(a) $A+B$

A	B	X
0	1	1
1	0	1
1	1	1
0	0	0

(b) AB

A	B	X
0	0	0
1	0	0
0	1	0
1	1	1

(c) $A + B(C + D)$

(d) $A + B + C$

(e) $AB + BC$

ABCDX
00000
10101
11000
01011
11111

ABCX
0000
1001
0101
0011
1101
0111
1111

ABCX
0000
1010
1101
0111
0100
1111

7. **[Section 4-4, Exercise 22]** Determine which of the logic circuits in Figure 4–59 are equivalent.

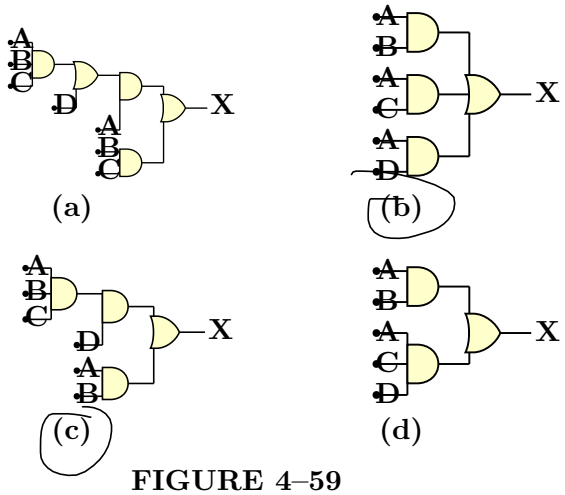


FIGURE 4–59

8. **[Section 4-7, Exercise 31]** Develop a truth table for each of the following standard SOP expressions:

(a) $ABC + \overline{A}BC + AB\overline{C}$

(b) $XYZ + X\overline{Y}Z + \overline{X}YZ + \overline{X}Y\overline{Z} + X\overline{Y}\overline{Z}$

ABCX
0000
0010
0100
0111
1000
1010
1101
1111

XYZ OUTPUT
000 0
001 0
010 1
011 1
100 0
101 1
110 1
111 1

9. **[Section 4-7, Exercise 36]** For each truth table in Table 4–15, derive a standard SOP and a standard POS expression.

ABC	X
0 0 0	0
0 0 1	1
0 1 0	0
0 1 1	0
1 0 0	1
1 0 1	1
1 1 0	0
1 1 1	1

(a)

ABC	X
0 0 0	0
0 0 1	0
0 1 0	0
0 1 1	0
1 0 0	0
1 0 1	1
1 1 0	1
1 1 1	1

(b)

$ABCD$	X
0 0 0 0	1
0 0 0 1	1
0 0 1 0	0
0 0 1 1	1
0 1 0 0	0
0 1 0 1	1
0 1 1 0	1
0 1 1 1	0
1 0 0 0	0
1 0 0 1	1
1 0 1 0	0
1 0 1 1	0
1 1 0 0	1
1 1 0 1	0
1 1 1 0	0
1 1 1 1	0

(c)

$ABCD$	X
0 0 0 0	0
0 0 0 1	0
0 0 1 0	1
0 0 1 1	0
0 1 0 0	1
0 1 0 1	1
0 1 1 0	0
0 1 1 1	1
1 0 0 0	0
1 0 0 1	0
1 0 1 0	0
1 0 1 1	1
1 1 0 0	1
1 1 0 1	0
1 1 1 0	0
1 1 1 1	1

(d)

TABLE 4–15

10. **[Section 4-9, Exercise 40]** Use a Karnaugh map to find the minimum SOP form for each expression:

(a) $\overline{A}\,\overline{B}C + A\overline{B}C + \overline{A}B\overline{C}$

(b) $AC(\overline{B} + C)$

(c) $\overline{A}(BC + B\overline{C}) + A(BC + B\overline{C})$

(d) $\overline{A}\,\overline{B}C + A\overline{B}\,\overline{C} + \overline{A}B\overline{C} + AB\overline{C}$

a. X = $\overline{B}C + \overline{A}B\overline{C}$

b. X = AC

c. X = B

d. X = $A\overline{C} + B\overline{C} + \overline{A}\overline{B}C = \overline{C}(A+B) + \overline{A}\overline{B}C$

a. X = $\overline{A}\overline{B}C + AB\overline{C} + A\overline{B}C + ABC$ X = (A + B + C)(A + $\overline{B}$ + C)(A + $\overline{B}$ + $\overline{C}$)($\overline{A}$ + $\overline{B}$ + C)

b. X = $A\overline{B}C + AB\overline{C} + ABC$ X = (A+B+C)(A+B+$\overline{C}$)(A+$\overline{B}$+C)(A+$\overline{B}$+$\overline{C}$)($\overline{A}$+B+C)

c. X = $\overline{A}\overline{B}C\overline{D}+\overline{A}\overline{B}CD+\overline{A}B\overline{C}D+\overline{A}BC\overline{D}+A\overline{B}C\overline{D}+A\overline{B}CD+AB\overline{C}\overline{D}$

d. X = $\overline{A}\overline{B}C\overline{D}+\overline{A}B\overline{C}\overline{D}+\overline{A}B\overline{C}D+\overline{A}BCD+A\overline{B}CD+AB\overline{C}\overline{D}+ABCD$